

NAMAN YESHWANTH KUMAR

linkedin.com/in/namany | github.com/naman-ranka | nyeshwan@asu.edu | (623) 249-1265

SUMMARY

MS Computer Engineering (ASU, GPA 3.81, May 2026) building production systems that process data at scale and integrate ML inference pipelines. Sole-built Trezzit (trezzit.com) — a multi-service production platform serving **600+ users** for 12+ months (PostgreSQL + Redis + Celery workers + multi-provider ML pipeline, **995 tests**, GitHub Actions CI/CD). Built SiliconCrew — an async job orchestration system with per-run isolation, exponential backoff, process-restart recovery, and GPU/CPU compute routing across 3 ML providers.

TECHNICAL SKILLS

Systems & Languages: Python | C++ | CUDA | Async Python | PostgreSQL | Redis | Celery | Docker | Linux | GitHub Actions CI/CD | Nginx | systemd

ML & Data Infrastructure: Multi-provider ML Routing | PyTorch | Data Pipelines | LLM Evaluation | Pydantic Validation | FastAPI | WebSocket | Streaming APIs

Compute & Architecture: GPU Programming (CUDA) | gem5 Architectural Simulation | Performance Profiling | Job Orchestration | Process Isolation | Exponential Backoff

EDUCATION

Master of Science — Computer Engineering (GPA 3.81/4.0)

Expected May 2026

Arizona State University

Tempe, AZ

Coursework: Artificial Intelligence · Reinforcement Learning in Robotics · Perception in Robotics · Foundations of Algorithms · Algorithm/Hardware Co-Design

Bachelor of Engineering — Electrical and Electronics

August 2022

BMS College of Engineering

Bengaluru, India

PROJECT EXPERIENCE

SiliconCrew — Async Job Orchestration & ML Inference Platform

Nov 2025 – Present

- Built an async job orchestration system (21 tools) routing ML inference across 3 providers (Gemini, OpenAI, Anthropic) with provider-specific failure handling and quality-based fallback; per-run process isolation, exponential backoff (30s→10min cap, 20-min hard timeout), and on-disk index enabling full job recovery after process restart; FastAPI + WebSocket backend with real-time streaming.
- Designed the data processing pipeline: ingests diagnostic output (simulation logs, waveform data, compiler reports), structures it for automated analysis, identifies root causes, and triggers re-processing — a closed-loop system that self-corrects without human intervention; achieved **37/92 first-pass** and **5/5 perfect score** on structured benchmarks.
- Built observability and monitoring: real-time WebSocket streaming of job execution traces, per-session compute cost tracking across all ML providers enabling cost-aware routing decisions, and pipeline-stage artifact inspection for debugging failed jobs without re-running.

Trezzit — Production AI SaaS (trezzit.com)

Feb 2025 – Present

- Sole-built a multi-service production platform serving **600+ users** for 12+ months — multi-provider ML inference pipeline (4 providers) with two-pass data processing and automatic fallback to higher-capability model on failure; O(1) pre-computed balance system with incremental updates, periodic reconciliation, and zero-sum invariant enforcement; SHA-256 deduplication for data ingestion.
- Production infrastructure: **995 tests** on every push via GitHub Actions CI with live PostgreSQL service container; Redis for caching and rate limiting; DigitalOcean VPS with Nginx + Gunicorn + systemd managing 4 services (web, Celery workers, Celery beat, monitoring); **60+ REST endpoints**, **40 DB models**; per-provider ML cost and latency tracking.

LRU-IPV Cache Replacement Policy — gem5 C++

Fall 2024

- Extended gem5's C++ architectural simulator with a new cache replacement policy as a first-class SimObject (4 interface methods across 5 source files); used controlled experiments across 3 L2 configurations (128/256/512 kB) to isolate capacity-vs-policy performance effects, correlating replacement behavior (637 → 112 → 2) with IPC at 41.8M instructions — demonstrating performance analysis methodology on large C++ codebases.

CUDA MLP — GPU Architecture Coursework

Fall 2024

- Implemented 6 CUDA kernels in C++ for MLP training (linear forward/backward, ReLU forward/backward) — manual GPU memory management (cudaMalloc/cudaMemcpy), kernel launch configuration tuned for occupancy; compiled via nvcc to shared libraries, bound to Python through ctypes FFI; end-to-end training pipeline from raw data to converged model on GPU.

WORK EXPERIENCE

Systems Engineer (Sole Developer)

Oct 2022 – Apr 2024

NCR GOLD

Bengaluru, India

- Automated the entire order lifecycle for a wholesale business — production Django platform (PostgreSQL + Celery + Nginx) processing **18,000+ orders over 3 years**; reverse-engineered third-party ERP (zero public API) via SQL schema analysis for bidirectional sync, eliminating ~20 min of manual relay per order.

AWARDS & HONORS

- Won **‘Best Use of AI’** at Strategy X DevHacks’25 (ASU) — built AdaptED AI, a Gemini-powered platform generating personalized flashcards, quizzes, and learning paths from uploaded documents in a 24-hour hackathon.
- Won **1st place at Devlabs Hackathon** — HEAL.AI, an AI healthcare assistant with a custom RAG pipeline for insurance Q&A and automated medical bill dispute generation.
- Secured **2nd place at Hack SoDA 2024** (Amazon-sponsored, 24-hour) — built PassGen, a secure offline Chrome Extension for password generation.